

Smart Student Expense Tracker

A lightweight, desktop-based offline spending logger built with Python and Tkinter. Designed specifically for university students to record daily expenditures, calculate totals, and perform index-based expense management.

1. PROJECT OVERVIEW & ARCHITECTURE

CORE OBJECTIVE

To provide an accessible, zero-overhead desktop application that simplifies daily financial tracking for college students without requiring database servers, cloud sign-ups, or active network connections.

ARCHITECTURE PATTERN

Event-Driven Modular GUI Architecture. User actions on Tkinter interface widgets trigger structured event handlers, maintaining state in a dynamic in-memory data store.

2. APPLICATION FEATURES

+ Add Expense

Records Date, Category, Description, and Amount in ₹ with field validation.

📄 Show Expenses

Displays a sequentially formatted list of all logged records with index numbers.

📊 Total Expense Sum

Calculates real-time arithmetic sum of all stored monetary amounts.

🗑️ Delete Expense

Removes a specific transaction item based on its list index number.

🧹 Clear All Expenses

Resets the tracker array completely, wiping all logged entries from memory.

⚠️ Input Validation

Generates interactive popup warnings for empty inputs or invalid bounds.

3. TECHNOLOGIES & TOOLS

Component	Technology / Module	Purpose & Usage
Language	Python 3.x	Core programming language and logic execution engine.
GUI Framework	Tkinter	Constructing desktop windows, labels, inputs, and buttons.
Dialog System	`tkinter.messagebox`	Rendering popup alerts, success dialogs, and output popups.
Version Control	Git & GitHub	Source code versioning, repository tracking, and submission.

4. DIRECTORY STRUCTURE

```
Smart-Student-Expense-Tracker/ | |— vit project.py # Main application source code file
                               | |— README.md # Primary GitHub documentation (this specification)
                               | |— statement.md # Problem statement & scope declaration
```

5. INSTALLATION & SETUP GUIDE

PREREQUISITES

Python 3.7 or higher installed on standard operating systems (Windows, macOS, or Linux). Tkinter is pre-bundled with standard Python installations on Windows and macOS.

1. Clone the Repository:

```
git clone https://github.com/your-username/Smart-Student-Expense-Tracker.git
```

2. Navigate to Project Folder:

```
cd Smart-Student-Expense-Tracker
```

3. Run the Application:

```
python "vit project.py"
```

6. QUALITY ASSURANCE & TESTING PROTOCOL

Execute the following manual test suite to verify full system functionality prior to deployment:

Test ID	Test Scenario	Action & Input	Expected Result	Status
TC_01	Validation Check	Leave fields empty & click Add Expense	Warning popup: "Please fill all the details"	PASSED
TC_02	Expense Creation	Enter: `2026-09-28`, `Food`, `Canteen`, `150`	Success popup & fields reset	PASSED
TC_03	List Formatting	Click Show Expenses button	Popup displaying itemized indexed text	PASSED
TC_04	Total Calculation	Click Total Expense button	Popup displaying cumulative sum in ₹	PASSED
TC_05	Invalid Deletion	Enter index `99` & click Delete Expense	Warning popup: "Invalid expense number"	PASSED
TC_06	Valid Deletion	Enter index `1` & click Delete Expense	Success popup & item removed	PASSED
TC_07	Bulk Clearance	Click Clear All Expenses	Success popup & array wiped	PASSED

7. GUI LAYOUT & INTERFACE MAPPING

INPUT FIELDS

- **Date Entry:** Text entry for transaction date.
- **Category Entry:** Text entry for expense type.
- **Description Entry:** Details of the transaction.
- **Amount Entry:** Numerical value in INR.

CONTROL ACTIONS

- **Add Expense:** Triggers `add\_expense()` handler.
- **Show Expenses:** Triggers `show\_expenses()` handler.
- **Total Expense:** Triggers `total\_expense()` handler.
- **Delete Expense:** Triggers `delete\_expense()` handler.